



REFERENCE

The smysl Document Format

Every construct of the .smy surface syntax, with a worked example for each

Vladimir Ulogov · 2026 · smysl 1.4.0 · format smysl/1.0 · kernel smysl.kernel/0.1

This reference covers one thing only: how to write a `.smy` document by hand — every record you can author, every field it takes, and the rules a validator will hold you to. For **why** the format is shaped this way, see `SMYSL_RATIONALE.typ`; this document only answers **how**. Every example below is real syntax, checked against the parser rather than invented for the page.

THIS GUIDE IS NOT THE CONTRACT

It is a **writer's** reference — descriptive, and free to explain more than it obliges. What another implementation must obey to interoperate is `SMYSL_FORMAT_SPEC.md`, which is normative and deliberately a fraction of this length: identity and how a uid is derived, the deterministic-CBOR constraints, record framing, the round-trip fixed point, rule X and the conformance classes. Nothing else is required.

RFC SMYSL-1, which earlier versions of this guide referred to, is retired. It was the product idea rather than the doctrine, and the implementation outgrew it.

How a document is put together

A `.smy` file is a flat sequence of **records**, each starting at column 0. Order does not matter to the meaning of the document — a claim can cite evidence that appears later in the file — though it is conventional to write things in the order a reader would want to meet them. Blank lines separate records but carry no meaning of their own.

COMMENTS

A line beginning `#` or `//` at column 0 is a comment. Both markers, because an HJSON header inside a record already accepted both, and rejecting between records what was accepted within one made the surface contradict itself.

A comment is a comment **wherever it sits**, including inside a body. The alternative was worse: a body runs from the gist to the next record, so a comment between two records fell inside that range and became the previous unit's body, inventing content out of a note.

A body that genuinely needs to open a line with a marker escapes it — `\#` and `\\//`, and `\\` for a line that already starts with a backslash. Only those three, and only at column 0, so prose full of Windows paths and LaTeX needs no thought. The writer puts the escape back, so `fmt` stays a fixed point.

New in 0.6. Before it there was no escape at all: a Markdown heading or a line of C++ in a body was read as a comment and dropped in silence, which is a poor way to treat exactly the content this format carries.

No record carries a comment, so canonical form cannot reproduce one and `fmt` warns before dropping them. A note that has to survive belongs in a unit — typically `@question` or `@prose` — where it is content and travels like content.

Inside a header the markers work the same way, which has one consequence for values: a value that **begins** with `#` or `//` must be quoted, or the rest of the line — closing brace included — is read as a comment. The writer quotes such a value for you. A marker in the **middle** of a value is ordinary text, because a quoteless value runs to `,`, `}`, `]` or end of line without stopping at either marker. So `ref: grafana://board/12` needs no quotes and gets none.

New in 0.2. Before that there was no comment syntax at all, and such a line was `SMY-E001`. The quoting rule for values is 0.4: until then the writer emitted such a value bare and lost the record it belonged to.

Any other line that is not part of a record's header, gist, body, or detail is a diagnostic (`SMY-E001`), not an ignored remark.

Four kinds of record are ever hand-authored in surface syntax:

CONSTRUCT	WHAT IT DECLARES
<code>@doc</code>	The document header — at most one per file, and optional.
a unit	<code>@claim</code> , <code>@evidence</code> , and the other thirteen kernel types — the thing being claimed, cited, or asked.
<code>@rel</code>	A typed edge between two units — <code>causes</code> , <code>rebutts</code> , <code>warrant</code> , and eleven more.
<code>@thread</code>	A named, ordered, role-annotated walk over units — a brief, a narrative, an analysis.
<code>@schema</code>	A declaration of an extension: the unit types and relation kinds it adds. Rarely needed, and one line when it is.

Other record kinds exist in the format and none is something you write directly. An **attestation** is stamped on automatically by whatever tool ingests or authors a unit; a **contention** is what `merge` produces when two documents disagree; a **pack manifest** is `pack`'s receipt; a **schema declaration** names an extension a store depends on.

A **label binding** is the one worth knowing about, because it explains a number. Writing `@claim c/cache-cold` declares both a unit and a name for it, and the name travels as its own record — so a labelled unit yields two. That is why `check` reports more records than you typed, and why `units` rather than `records` is the figure to read when you want to know how much document you have.

The binding has to be separate because a label is **not identity**: it is not hashed, and renaming one must not produce a different unit. New in 0.2 — before it, labels survived a parse and not a store round trip, so a document that had been through `merge` came back with every reference spelled as a bare uid.

The `@doc` header

If present, `@doc` must be the first thing in the file, and it fixes the document's own identity and the terms under which everything else in it should be read:

```
@doc smysl/0.1 {
  id: v/f1
  intent: incident-brief
  lang: en
  requires: ["smysl.kernel/0.1"]
  granularity: { profile: default }
  roots: [f/root-cause]
}
```

FIELD	DEFAULT	MEANING
<code>id</code>	<code>v/doc</code>	This document's own identifier — a view id, <code>v/<slug></code> .
<code>intent</code>	empty	A free-text label for what the document is for, e.g. <code>incident-brief</code> .
<code>lang</code>	<code>en</code>	A BCP-47 language tag — up to 35 characters, hyphen-separated alphanumeric segments.
<code>requires</code>	empty	Schemas a full-fidelity reader must implement: the kernel schema itself and any extensions.
<code>roots</code>	empty	The units this view is about — everything reachable from them is in the view; nothing is copied.
<code>threads</code>	empty	Thread ids this view publishes, if any.
<code>granularity</code>	<code>default</code> profile	How much one unit is allowed to say — see below.

A view is not a container: `roots` names starting points, and everything reachable from them via `deps`, `grounds`, and relation edges belongs to the view at zero copying cost. That is also what makes merging two documents a plain union rather than a conflict.

Granularity

`granularity` bounds how much a single unit is allowed to say, as three presets or a custom mix. Granularity constrains **production** — what you should write — not the store: a merged corpus mixing granularities is legal, not an error.

PROFILE	GIST (L0)	BODY (L1)	ADMISSION
<code>coarse</code>	≤ 30 tokens	120 – 400 tokens	<code>topical</code> — one unit may bundle a topic
<code>default</code>	≤ 30 tokens	40 – 120 tokens	<code>single-assertion</code> — one unit, one claim

PROFILE	GIST (L0)	BODY (L1)	ADMISSION
<code>fine</code>	≤ 30 tokens	20 – 60 tokens	<code>single-assertion</code>

Under `single-assertion` admission — the default — a body that bundles more than one claim into a single unit is `SMY-E040`, an error, not a style complaint: split it into two units and a relation between them instead. A body outside its profile's range is only `SMY-W041`, a warning.

Units — the shared anatomy

Every unit, whatever its type, shares the same header and the same three-level text. Here is the fullest ordinary example the format has:

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait], salience:
0.8 }
~ The eu-west connection pool is saturated.
  Waits climbed from a steady 2 ms baseline to over 300 ms in the same hour the 4.2
  rollout reached eu-west, and no other shard's pool moved.

--
Per-shard breakdown at hourly resolution: eu-west crosses 250 ms at 09:14 UTC,
forty
minutes after the rollout's first canary step there. No other shard's pool exceeds
its usual 5 ms band over the same seven-day window.
```

FIELD	REQUIRED WHEN	WHAT IT IS
label	optional	The word right after the type, before <code>{</code> , e.g. <code>c/pool-saturation</code> — how the rest of the file refers to this unit. Not identity: two labels never make two units, and a unit needs no label at all if nothing else in the file must reference it by name.
<code>status</code>	always — defaults to <code>speculative</code>	How sure this unit is entitled to be — the six-rung ladder, next section.
gist (<code>~ ...</code>)	always	The one required line of text — L0. Must sit on the line immediately after the header, with no blank line between.
body	needed by <code>derived</code> / <code>inferred</code> gists that lean on it	L1 — a paragraph or two, the ordinary prose account.
detail	only after a body	L2 — unbounded, introduced by a lone <code>--</code> line. A detail with no body above it is <code>SMY-E023</code> .
<code>deps</code>	when the gist needs context to parse	Other units this one's wording depends on to be understood — not evidence, just prerequisite reading.
<code>grounds</code>	required by <code>derived</code> / <code>inferred</code>	Other units this one's truth rests on. Retract one and this unit's status is re-opened.

FIELD	REQUIRED WHEN	WHAT IT IS
<code>source</code>	required by <code>measured</code> / <code>cited</code>	Where this unit grounds out externally — see below.
<code>salience</code>	optional	An authored override in <code>[0,1]</code> , clamped and quantised to 1/1024. Left unset, salience is computed rather than declared.

The gist must immediately follow the header — not even a blank line may separate them (SMY-E021). A gist continuation line is indented by exactly two spaces; a body or detail line is not indented at all. The assembled gist is trimmed: whitespace around it is never content, because the reader strips the `~` sigil and the space after it, so a gist that carried a leading space could not be written back and read again unchanged.

WHAT IS NOT CONTENT

Identity is content, so anything the format treats as **not** content has to be settled before a uid is computed — otherwise two documents that say the same thing hash differently. Three things are normalised on the way in:

- **Line endings.** A CRLF file and an LF file are the same document. Every carriage return at the end of a line is stripped, not just one. A carriage return **inside** a line is ordinary text and survives.
- **Whitespace around a gist,** as above.
- **Unicode form.** All text is normalised to NFC — including the unknown header keys and values that rule X carries verbatim. `é` written as one code point and `e` plus a combining accent are the same text and get the same uid.
- **A repeated field.** Writing `deps` twice in one header keeps the first and discards the rest. Before 0.5 the second was carried as an unknown key under rule X and written back out as a plain `deps:` line, which the next parse then read as the field — so the document lost content by being reformatted. There is no way to spell a second `deps` in surface syntax, so first-wins is the only rule that round-trips.

Each of these was a real defect before 0.4, and the Unicode one was the worst of them: it failed silently in release builds, so two peers could write identical content and disagree about its identity.

Status: the six-rung ladder

STATUS	NEEDS	MEANING
<code>unfounded</code>	—	Reachable only by retraction; MUST NOT be authored (SMY-E034).
<code>speculative</code>	neither	Offered, not yet grounded — the default if you omit <code>status</code> .
<code>inferred</code>	<code>grounds</code>	A model reasoned it out from other units.
<code>derived</code>	<code>grounds</code>	Computed from other units by a deterministic procedure.
<code>cited</code>	<code>source</code>	A source you can open says so.
<code>measured</code>	<code>source</code>	An instrument recorded it.

Two mechanical rules sit on top of this table. **Rule M:** a unit's status can never outrank the weakest thing in its own `grounds` (SMY-E030) — a `derived` claim resting on a `speculative` one is itself no better

than speculative, whatever status you write. **Rule T:** the **rung** an agent is working from caps what status it may even attempt:

RUNG	CEILING	WHAT IT IS
computed	derived	A deterministic tool, calculation, or parser.
document	cited	A user-supplied document or dataset.
web	cited	Fetch content, gated.
model	inferred	The model's own parametric knowledge — never higher, however confidently phrased.

No rung reaches **measured** on its own, and the way it **is** reached is worth stating exactly, because the obvious reading is wrong. It takes an attestation recording **op: imported** *at the* **computed** *rung* — a deterministic tool transcribing a reading. The op alone is not enough: **ingest** also records **imported**, because it too transcribes rather than authors, so unlocking on the op by itself would let a model mark anything it read in a document as measured. Ingest runs at **document**, **web** or **model** and stays capped there.

WHY YOU CAN STILL TYPE IT

You may write **status: measured** in a **.smy** file and **check** will pass it, which looks like a contradiction and is not. Rule T is checked against a unit's **attestation**, and a **.smy** file cannot express one — provenance has no surface syntax. With nothing recorded about where a unit came from, there is no ceiling to check it against, so the rule stands aside rather than guessing. What you have written is an unchecked claim, not a licensed one, and it becomes checkable the moment the unit acquires provenance.

The producer that writes **measured** **with** the provenance that licenses it is the **import** command, which transcribes a table of readings. A unit whose attestation records some other op is **SMY-W035**.

Keys the grammar does not know

A header key that is not one of the fields above is **not** an error. It is kept, verbatim, in the unit's payload and written back out in the same place — the surface half of rule X. A reader from a later version, or from a team that records something yours does not, loses nothing by passing through yours.

```
@data t/latency { status: measured, source: { kind: file, ref: "by-region.csv" },
                  x.stats/method: "empirical-quantile",
                  columns: ["region", "p50_ms", "p95_ms"], rows: 6 }
~ Checkout latency by region, June 2026.
```

rows, **columns** and **x.stats/method** are none of the kernel's business. They survive a parse, a round trip through the binary encoding, and a merge — and they come back in canonical order rather than the order you typed them, which is why **smy** **fmt** returns the unit above with **rows** first and **x.stats/method** last. Keys sort by encoded length, then content. That is not tidiness: identity is computed from the encoded bytes, so two people who record the same thing in a different order still compute the same unit.

TWO NAMES THE TOOLING WRITES

Ingest puts two of its own keys in the same place, and you will meet them reading a staged file rather than writing one. `ingest:quote` carries the span of the source document a unit was drawn from — checked against that document, so a quote that is not in it is an error rather than a decoration. `ingest:unrepaired` marks a span that survived its repair budget and was kept as opaque prose rather than dropped. Neither is reserved by the grammar; both are conventions you can read, write, or ignore.

Source references

```
source: { kind: metric, ref: "pool.wait_ms{shard=eu-west}", captured: 2026-07-09 }
```

`kind` is one of five, `ref` (or the longer spelling `reference`) is a free-text pointer whose shape depends on the kind, and `captured` is an optional `YYYY-MM-DD` date.

KIND	TYPICAL REF	CAPTURE DATE
<code>url</code>	a web address	Required — a fetched page is unverifiable later without one.
<code>file</code>	a path or filename	optional
<code>metric</code>	a metric name or query	optional
<code>tool</code>	a tool or command name	optional
<code>doc</code>	a document reference or anchor	optional

The thirteen unit types

`KernelType` actually enumerates fifteen names, but two of them — `contention` and `packinfo` — name records that tooling produces (`merge`, `pack`); nothing below shows them being hand-authored, because in ordinary use they never are. The other thirteen are yours to write. Each example is complete and independently valid.

TYPE	WHAT IT RECORDS
<code>claim</code>	An assertion about the world.
<code>evidence</code>	A reading, instrument, or observation offered to ground a claim.
<code>definition</code>	What a term means, fixed so later claims can rely on it.
<code>question</code>	An open prompt the corpus has not yet answered.
<code>hypothesis</code>	A candidate answer, offered before it is tested.
<code>finding</code>	A conclusion resting on other claims.
<code>procedure</code>	A course of action — a “what to do,” not a “what is.”
<code>decision</code>	A commitment a person or team made.
<code>constraint</code>	A rule the rest of the reasoning has to respect.
<code>observation</code>	A plain record of what was seen.
<code>data</code>	A pointer to a dataset, rather than one reading.
<code>artifact-ref</code>	A pointer to something outside the document — a dashboard, a repo.

TYPE	WHAT IT RECORDS
prose	Free-standing text that fits none of the above — still a first-class, citable unit.

```
@evidence e/pool-wait { status: measured, source: { kind: metric, ref:
"pool.wait_ms{shard=eu-west}", captured: 2026-07-09 } }
~ Pool acquisition wait rose from 2 ms to 310 ms over the same window.
```

An instrument reading grounds out externally, so **measured** demands a **source** — here a metric reference with the date it was captured.

```
@claim c/pool-saturation { status: inferred, grounds: [e/pool-wait] }
~ The eu-west connection pool is saturated.
```

A model reasoned this from the evidence above; **inferred** demands **grounds**, and rule M means this claim can never outrank **e/pool-wait** even if the wording sounds certain.

```
@definition d/p95 { status: cited, source: { kind: doc, ref: "sre-
handbook#latency" } }
~ The 95th percentile of request latency over a one-minute window.
```

A definition is grounded in a document a reader can open, not the author’s own say-so — without it, “tripled” in a later claim would have no fixed meaning.

```
@question q/root-cause { status: speculative }
~ What actually caused the eu-west latency spike?
```

A question needs neither **source** nor **grounds** — **speculative** is the floor, an open prompt nothing has answered yet.

```
@hypothesis h/pool-exhaustion { status: speculative }
~ Connection-pool exhaustion under the 4.2 rollout is driving the p95 rise.
```

A candidate answer to the question above, offered before it is tested against evidence.

```
@finding f/root-cause { status: inferred, grounds: [c/pool-saturation, c/
regression] }
~ Pool saturation is the leading cause but is not consistent with the canary.
```

A finding rests on other claims rather than on raw evidence; rule M caps it at the weakest of the two grounds it names.

```
@procedure p/rollback { status: speculative }
~ Roll the eu-west pool config back to the 4.1 settings and re-run the canary.
```

A course of action, not a fact — still a checkable unit, just one whose content is a “what to do.”

```
@decision de/rollback-approved { status: inferred, grounds: [f/root-cause] }
~ Approved: roll back the eu-west pool config tonight.
```

A commitment, grounded in whatever finding justified it — retract the finding and this decision's grounding is reopened along with it.

```
@constraint co/no-friday-deploys { status: cited, source: { kind: doc, ref: "ops-
runbook#change-freeze" } }
~ No production rollbacks after 16:00 on a Friday.
```

A rule the rest of the reasoning has to respect, sourced from wherever that rule is actually written down.

```
@observation ob/canary-quiet { status: measured, source: { kind: metric, ref:
"canary.p95" } }
~ The 4.2 canary showed no latency regression over the same window.
```

A plain record of what was seen — it doesn't ground anything by itself yet, but it can be cited later, exactly like any other unit.

```
@data da/latency-series { status: measured, source: { kind: file, ref: "auth-
p95-2026-07.csv" } }
~ Hourly p95 auth latency, all shards, 2026-07-02 through 2026-07-09.
```

Points at a dataset rather than asserting a single reading — the source is the file itself.

```
@artifact-ref ar/dashboard { status: cited, source: { kind: url, ref: "https://
grafana.internal/d/api-latency", captured: 2026-07-09 } }
~ The latency dashboard the on-call team watches.
```

Points at something that exists outside the document. Its source kind is `url`, the one kind where `captured` is mandatory rather than optional.

```
@prose pr/note { status: speculative }
~ Worth a follow-up: check whether the 4.1 pool size was ever tuned for eu-west's
traffic shape.
```

The escape hatch — a citable aside that doesn't fit any more specific type.

NAMESPACE CONVENTION, NOT GRAMMAR

`e/`, `c/`, `d/`, `f/`, and the rest above are a **habit**, not a rule. A label is any `ident/ident` pair — lowercase letters, digits, `-`, `_` — and nothing in the grammar reserves any prefix for any unit type. Pick short, memorable prefixes and stay consistent within a file; the parser does not care what they are.

Relations (`@rel`)

A relation is a typed, directed edge between two units, written on one line:

```
<from> --<kind>--> <to> [{ weight: <0..1>, note: <ref> }]
```

```
@rel c/canary-clean --rebutts--> c/pool-saturation { weight: 0.6 }
```

A direct objection. `rebutts` is the one kind rule R pins: wherever `c/pool-saturation` survives into a packed or rendered view, this edge — and the unit it points from — travels with it. `weight` is an optional strength for the rebuttal itself, clamped to `[0,1]`.

```
@rel c/pool-saturation --causes--> c/regression
```

Says the saturation claim is what produced the regression. `causes` is one of three **ordering** kinds (with `enables` and `sequences`) that a thread’s automatic ordering can sort by, and one of two **support-bearing** kinds (with `answers`) that carry salience rank forward through the graph.

```
@rel d/p95 --warrant--> c/regression
```

Says the definition licenses reading the claim the way it is read — without it, “tripled” has no fixed meaning to check.

```
@rel c/pool-saturation-v2 --supersedes--> c/pool-saturation
```

Marks the newer claim as replacing the older one outright. `supersedes` and `retracts` are the two **lifecycle** kinds — the only ones that change what the corpus should currently be read as believing, rather than adding a new relationship alongside what was already there.

```
@rel c/pool-saturation --x.sre/mitigates--> p/rollback
```

An unrecognised kind — anything shaped `x.<domain>/<kind>` — is kept and stays routable rather than dropped (SMY-W013); a reader without the `x.sre` extension treats it as a plain `elaborates` edge instead of losing it.

The warning goes away when the file declares the kind it uses:

```
@schema x.sre/v1 { version: 1, relations: [x.sre/mitigates] }
```

The declaration travels with the store through `merge` and `fmt`. A misspelled key in it is an error rather than something skipped — a `relation:` quietly ignored would leave the kind warning under a file that appears to declare it — and declaring a kernel kind such as `causes` is SMY-E012, because an extension may add but never redefine.

The fourteen kernel kinds, for reference:

KIND	READS AS
<code>elaborates</code>	adds detail without changing the claim
<code>contrasts</code>	sets two things against each other
<code>concedes</code>	grants a point while maintaining the larger claim
<code>causes</code>	one thing produced another (ordering , support-bearing)
<code>enables</code>	one thing made another possible without directly producing it (ordering)
<code>exemplifies</code>	gives a concrete instance of a general claim
<code>conditions</code>	states an “only if” dependency
<code>sequences</code>	orders two things in time, with no causal claim (ordering)

KIND	READS AS
answers	resolves a question (support-bearing)
rebutts	directly objects to the target — pinned by rule R
warrant	licenses an inference or a reading
backs	supports a warrant
supersedes	replaces the target outright (lifecycle)
retracts	withdraws the target outright (lifecycle)

Threads (`@thread`)

A thread is a named, ordered, role-annotated walk over units — the same units the corpus already has, arranged into a reading order for one audience. Its header takes a **schema** (which fixes the allowed roles), an **owner** (an agent id — a thread is **owned** state, last- writer-wins **per owner**, so two agents publishing the same thread id never conflict), and a gist. Steps follow, indented by two spaces:

```
<role> → <unit>[: <note>]
```

Each of the five schemas fixes a different sequence of roles. One worked example per schema:

```
@thread t/brief { schema: brief, owner: "human:vladimir" }
~ Auth p95 tripled in eu-west; pool saturation is leading but contested.
  bottom-line → f/root-cause
  support → c/pool-saturation
  risk → c/canary-clean
```

brief — **bottom-line**, **support**, **risk**, **ask** — walks a reader from the one-line verdict down to what backs it and what argues against it. Not every role has to be used.

```
@thread t/root-cause-qa { schema: qa, owner: "human:vladimir" }
~ What caused the latency spike, and how sure are we?
  question → q/root-cause
  evidence → e/pool-wait
  answer → f/root-cause
  caveat → c/canary-clean
```

qa — **question**, **evidence**, **answer**, **caveat** — is literally a question answered in public, with its evidence and its own best objection attached.

```
@thread t/incident-story { schema: narrative, owner: "human:vladimir" }
~ How a quiet Tuesday became an eu-west incident.
  setup → ob/canary-quiet
  complication → e/pool-wait
  turn → c/pool-saturation
  resolution → de/rollback-approved
  coda → f/root-cause
```

narrative — **setup**, **complication**, **turn**, **resolution**, **coda** — orders the same kind of units as a story, for an audience that needs to follow how the conclusion was reached.

```
@thread t/deep-dive { schema: analysis, owner: "model:anthropic/claude-opus-5" }
~ Full trace of the eu-west pool-saturation analysis.
  context → ob/canary-quiet
  tension → e/pool-wait
  approach → h/pool-exhaustion
  finding → c/pool-saturation
  rebuttal → c/canary-clean
  implication → f/root-cause
  next → p/rollback
```

analysis — **context**, **tension**, **approach**, **finding**, **rebuttal**, **implication**, **next** — is the longest schema, a full research trace with its own slot for what to do next.

```
@thread t/rollback-plan { schema: plan, owner: "human:vladimir" }
~ Rolling eu-west back to 4.1 tonight.
  goal → de/rollback-approved
  constraint → co/no-friday-deploys
  step → p/rollback
  risk → c/canary-clean
```

plan — **goal**, **constraint**, **step**, **decision**, **risk** — orders the units that make up a course of action rather than an argument.

ARROWS AND NOTES

Either → (U+2192) or the ASCII -> is accepted when **reading**; the canonical writer (`smysl fmt`) always emits →. A step may carry a trailing **: note** after the reference, as free text — support → c/pool-saturation: the strongest single reading.

risk appears in both **brief** and **plan** — it is one role shared between schemas, not two separate ones. A step whose role the thread's schema does not declare is not rejected; it is reported as a **foreign role**, kept rather than dropped.

Identifiers, at a glance

KIND	SHAPE	EXAMPLE
Label	<code>ident/ident</code> , lowercase	<code>c/pool-saturation</code>
Thread / view / contention id	same shape, label-typed	<code>t/brief</code> , <code>v/f1</code> , <code>k/pool-vs-index</code>
Agent id	<code>(model\ human\ tool):name[/tag]</code>	<code>model:anthropic/claude-opus-5</code> , <code>human:vladimir</code>
Kernel type	a bare word	<code>claim</code> , <code>evidence</code>
Kernel schema	<code>smysl.kernel/MAJOR[.MINOR]</code>	<code>smysl.kernel/0.1</code>
Extension schema	<code>x.<domain>/<segment></code>	<code>x.sre/incident</code>
Language tag	BCP-47-shaped, ≤ 35 chars	<code>en</code> , <code>en-GB</code> , <code>zh-Hans-CN</code>

KIND	SHAPE	EXAMPLE
Content hash (uid)	b3: + base32, machine-facing	b3: cvhirtgs2mpvli2ethhyeo32uf...

You will essentially never type a **b3:** hash by hand — that is exactly what labels are for. A hash is what a unit's identity actually is; a label is the name you gave it so you never have to write the hash yourself. An agent id's kind is fixed to one of three words; a model id's segment after the first **/** groups corroboration by provider, which is why **model:anthropic/claude-opus-5** and **model:anthropic/claude-haiku-4-5** still corroborate as the same provider.

Validation, at a glance

A parse never simply fails: a malformed record becomes a diagnostic with a byte span, and parsing recovers at the next record start, so one bad unit costs you that unit, not the whole file. The codes below are the ones ordinary hand-authoring runs into.

CODE	WHAT IT MEANS
SMY-E001	General surface parse error — a malformed header, an unknown type, a stray line.
SMY-E020	Body or detail names a unit's identifier that isn't listed in deps or grounds .
SMY-E021	Missing gist, or the gist doesn't immediately follow the header.
SMY-E022	Gist exceeds the profile's l0_max .
SMY-E023	detail present without a body above it.
SMY-E030	Rule M — status exceeds the weakest of its own grounds.
SMY-E031	derived / inferred with empty grounds .
SMY-E032	measured / cited without a source .
SMY-E033	Rule T — status exceeds the ceiling for the authoring rung.
SMY-E034	unfounded authored directly — only reachable by retraction.
SMY-E040	More than one assertion in a body under single-assertion admission.
SMY-E060	A label referenced but never defined anywhere in the document.
SMY-E061	A cycle in deps or grounds .
SMY-W013	Unknown relation kind — kept, treated as elaborates for closure.
SMY-W035	measured on an authored (not imported) unit.
SMY-W041	Body length outside the profile's l1_range — advisory only.

Three more you will only meet in a document a model produced, never one you typed. They are listed because you will read them in a staged file and should know they are not your mistake:

CODE	WHAT IT MEANS
SMY-W036	Rule M applied at ingest — a unit claimed more than its grounds support and was lowered to what they do. The record of a model overreaching, not work outstanding.
SMY-E307	An attributed ingest:quote does not occur in the source document. A fabricated attribution, and the one thing here worth another turn of the model.
SMY-W308	The quote occurs only loosely — elided or reworded. Honest attribution with a clause dropped.

AT A GLANCE

- Four constructs are hand-authored: `@doc` once, units as many as you need, `@rel` for typed edges, `@thread` to walk them in an order.
- Every unit shares one anatomy — label, status, gist/body/detail, `deps/grounds`, `source`, `salience` — whatever its type.
- Status only ever falls, never rises, across `grounds` (rule M) and is ceilinged by the rung doing the authoring (rule T); `measured` needs `op: imported` at the `computed` rung, which in practice means `smysl import` and nothing else.
- Thirteen unit types are yours to write; `contention` and `packinfo` are tooling-only.
- Fourteen relation kinds, five thread schemas, twenty-four roles — all closed sets, all with an escape hatch (`x.<domain>/<kind>`) that degrades gracefully rather than being dropped.
- Header keys the grammar does not know are kept verbatim in the payload rather than refused, so a document from a later version or another team passes through yours intact.
- A `#` or `//` line at column 0 is a comment, and no record carries one — `fmt` warns before dropping them. There is still no blank line permitted between a header and its gist, and no reserved label namespaces: only what the grammar actually checks.
- A labelled unit yields two records, because the label travels as its own binding rather than inside the hashed content that decides identity.